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Contents

Intellectual Property Rights 2
Foreword 2
Foreword 4
1 Scope 5
2 References 5
3 Abbreviations 5
4 C code structure 5
4.1 Contents of the C source code 6
4.2 Program execution 7
4.3 Variables, constants and tables 9
4.3.1 Description of constants used in the C-code 9
4.3.2 Type Definitions 10
4.3.3 Description of fixed tables used in the C-code 13
4.3.4 Static variables used in the C-code 14
4.4 Functions of the C Code 14
4.4.1 Interface functions 15
4.4.2 IVS transmitter functions 17
4.4.3 PSAP receiver functions 18
4.4.4 PSAP transmitter functions 21
4.4.5 IVS receiver functions 21
4.4.6 Synchronization functions (IVS and PSAP) 21
4.4.7 Control link functions 23
4.4.8 Other utility functions (IVS and PSAP) 24
Annex A (informative): Change history 25
History 26

Foreword

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1 Scope

The present document contains an electronic copy of the ANSI-C code for the eCall in-band modem solution for reliable transmission of MSD data from IVS to PSAP via the speech channel of cellular networks. The ANSI-C code is necessary for a bit exact implementation of the IVS modem and PSAP modem described in 3GPP TS 26.267 [1].

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

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[1] 3GPP TS 26.267: "eCall Data Transfer; In-band modem solution; General description".

See also the references in 3GPP TS 26.267 [1].

3 Abbreviations

For the purpose of the present document, the following abbreviations apply:

ACK	ACKnowledgement
ANSI	American National Standards Institute
CRC	Cyclic Redundancy Check
FEC	Forward Error Correction
GSM	Global System for Mobile communications
HARQ	Hybrid Automatic Repeat-reQuest
I/O	Input/Output
IVS	In-Vehicle System
MSD	Minimum Set of Data
NACK	Negative ACKnowledgement
PCM	Pulse Code Modulation
PSAP	Public Safety Answering Point
RAM	Random Access Memory
ROM	Read Only Memory
RX	Receive
TX	Transmit

4 C code structure

This clause gives an overview of the structure of the bit-exact C code and provides an overview of the contents and organization of the C code attached to the present document.

The C code has been verified on the following systems:

- Windows XP SP2 and Microsoft Visual Studio V8.0;
- Linux (Suse Linux) using the gcc v3.4.2 and v4.1.2 compilers.

4.1 Contents of the C source code

The distributed files with suffix "c" contain the source code and the files with suffix "h" are the header files.

Further explanation on the files is given in the `readme.txt` file, which is reproduced in part here:

```
Package Contents
----------------

folder 'ecall':
    Contains the complete eCall ANSI C fixed-point reference source code.

    modem_ivs.c     : top-level modem implementation for IVS
    modem_psap.c    : top-level modem implementation for PSAP

    modemx.h        : header file for both modem_ivs.c and modem_psap.c

    ecall_defines.h : compile time options and preprocessor constants

    ecall_control.h : header file control message handling
    ecall_fec.h     : header file FEC encoder and decoder
    ecall_modem.h   : header file modulator and demodulator
    ecall_sync.h    : header file synchronization
    ecall_rom.h     : header file ROM data

    ecall_control.c : control message handling
    ecall_fec.c     : FEC encoder and decoder
    ecall_modem.c   : modulator and demodulator
    ecall_sync.c    : synchronization
    ecall_rom.c     : ROM data


folder 'test_setup':
    Contains the eCall software simulation framework, to be compiled
    and run on MS Windows systems.

folder 'test_vec':
    Contains binary PCM data (104 files) and receiver/transmitter port logs
    in ASCII format (104 files) to test the eCall IVS and PSAP modems.

    The PCM format is 16 bit signed, little endian, at 8 kHz sampling rate.
    The data files reflect 26 test cases and were generated from the eCall
    simulation framework.

    campaign_short.txt    : configuration file for the 26 test cases

    pcmdlout<index>.pcm   : output PCM data of DL vocoder = input to IVS
    pcmulout<index>.pcm   : output PCM data of UL vocoder = input to PSAP

    pcmdlin<index>.pcm    : test vectors for PSAP modem output
    pcmulin<index>.pcm    : test vectors for IVS modem output

    portivsrx<index>.txt  : test vectors for IVS port logs (receiver)
    portivstx<index>.txt  : test vectors for IVS port logs (transmitter)

    portpsaprx<index>.txt : test vectors for PSAP port logs (receiver)
    portpsaptx<index>.txt : test vectors for PSAP port logs (transmitter)

standalone.c
    main() wrapper to run the IVS or PSAP modem on prestored PCM files or
    receiver/transmitter port logs. To get a list of command-line options,
    invoke the corresponding executable with option '-h' (help).

standalone.h
    header file for standalone.c


Makefile.win
```

```
    Microsoft Visual Studio 2005/2008 Makefile
    Builds 'standalone.exe' from standalone.c and the eCall sources,
    build options are RELEASE and DEBUG.

Makefile.glx
    GNU Linux Makefile using gcc
    Builds 'standalone' from standalone.c and the eCall sources,
    build options are RELEASE and DEBUG.

verify.bat
    Windows batch file
    Runs 'standalone.exe' in six different modem modes on the 26 test cases
    contained in folder 'test_vec' and performs a test vector comparison to
    the respective output PCM and port log data.

verify.sh
    Linux shell script
    Runs 'standalone' in mode '-m ivs' and '-m psap' on 26 test cases
    (folder 'pcm') and performs a test vector comparison to the respective
    modem output PCM data.
```

4.2 Program execution

An explanation on code compilation and execution is given in the `readme.txt` file, which is reproduced in part here:

```
Getting Started
---------------

3GPP TS 26.268 provides the eCall modem source code, a software simulation
framework, and a standalone wrapper that allows to run the IVS or PSAP modem
on prestored reference data.

The following functions represent the eCall modem interface and invoke the
respective receiver and transmitter implementation of each modem:

    * void PsapReset();
    * void PsapProcess(Int16 *pcm);
    * void PsapSendStart();
    * void PsapSendHlack(const Ord8 data);


    * void IvsReset(const Ord8 *msd, int length);
    * void IvsProcess(Int16 *pcm);
    * void IvsSendStart();

The external application must in addition implement the callback functions:

    * void PsapReceiveMsd(const Ord8 *msd, int length);
    * void IvsReceiveAck();
    * void IvsReceiveHlack(const Ord8 data);

    * void Abort(const char *format, ...);
    * void LogInfo(const char *format, ...);


They will be called
    a) by the PSAP modem once the complete MSD was successfully received,
    b) by the IVS modem on reception of the lower-layer ACK,
    c) by the IVS modem on reception of the HLACK message.

Abort and LogInfo should implement a variadic error and printlog handler,
respectively. See standalone.c for sample implementations of all callback
functions.

For a real-time simulation over 3GPP FR and AMR vocoders and to log PCM data
```

```
as input to the standalone wrapper, the eCall sources have to be integrated into a
simulation framework; folder 'test_setup' contains the one as used in
the 3GPP selection tests. The basic integration steps are briefly described
below.

In order to compile and run the eCall modem code, follow the instructions
given below. For code testing, two batch files have been provided:

    * verify.bat : MS Windows systems
    * verify.sh  : Linux systems

For each of the 26 test cases of campaign_short.txt in folder 'test_vec',
they run the standalone wrapper in six different modem modes (three IVS and
three PSAP modes). The resulting PCM and port log files in folder 'out' are
finally compared to the test vectors in folder 'test_vec'.

In modes 'psap' and 'psaprx', you should see an MSD success message at the
end of each test case.


Code Compilation
----------------

MS Windows systems
    To build standalone.exe from standalone.c and the eCall sources, start
    with opening a new project in Visual Studio 2005/2008.

    Choose File -> New -> 'Project from Existing Code' and follow the
    instructions of the 'Create Project from Existing Code Files Wizard'.
    Configuration:

    * Type of project: Visual C++
    * Specify the folder location of standalone.c and a project name
    * Button 'Next'
    * Select 'Use external build system'
    * For Debug and Release configuration, specify

    Build command line:  nmake -f Makefile.win
    Clean command line:  nmake -f Makefile.win clean

    Build the project with shorthand key 'F7' or from the menu.
    The source code should compile without any errors or warnings.
    Run 'verify.bat' to verify the executable against the test vectors.


GNU Linux systems
    Compilation under Linux has been tested with

    * GNU Make version 3.81
    * gcc version 4.1.3 and 4.2.4

    For building the executable 'standalone' and cleanup, use

    make -f Makefile.glx
    make -f Makefile.glx clean

    On the platforms tested, the code compiled without errors or warnings.
    Run 'verify.sh' to verify the executable against the test vectors.


Simulation Framework
--------------------

The eCall software simulation framework is provided in folder 'test_setup'.

Important remarks:

    * See LICENSE.TXT and README.TXT for terms of usage!

    * The G.711 software is part of ITU-T Rec. G.191, (C) ITU 2000.
```

```
      Distributed with the authorization of ITU as part of the test setup
      software for 3GPP TS 26.268.

    * The framework must be compiled and run on MS Windows systems, as the
      FR and AMR vocoders are attached to it in form of Windows executables
      and via Windows specific API functions.

To attach the eCall sources to the framework:

    * Copy the 'ecall' folder into subfolder 'c' of 'test_setup'.
    * Compile and link the *.c files under subfolder 'ecall' by adding their
      corresponding object files to the list of makefile targets.
    * In the makefile, replace the source code template modem_demo.c by both
      modem_ivs.c and modem_psap.c.

The framework has the callback functions PsapReceiveMsd, Abort, and LogInfo
already implemented.
```

4.3 Variables, constants and tables

4.3.1 Description of constants used in the C-code

This clause contains a listing of all global constants defined in `ecall_defines.h`., together with some explanatory comments.

Constant	Value	Description
`#define MAX(a,b)`	`((a)>(b) ? (a) : (b))`	
`#define MIN(a,b)`	`((a)<(b) ? (a) : (b))`	
`#define ABS(a)`	`((a)<0 ? (-a) : (a))`	
`#define SIGN(a)`	`((a)<0 ? (-1) : (1))`	
`#define PCM_LENGTH`	`160`	length of PCM frame
`#define MSD_MAX_LENGTH`	`140`	length of MSD message (bytes)
`/* Synchronization */`		
`#define SYNC_BADCHECKS`	`(8)`	IVS subsequent bad checks
`#define SYNC_IDXLEN`	`(75)`	sync index length
`#define SYNC_THRESHOLD`	`(10e6)`	sync threshold
`#define LOCK_START_UL`	`(2)`	START messages to lock sync (UL)
`#define LOCK_START_DL`	`(3)`	START messages to lock sync (DL)
`#define FAIL_RESTART`	`(3)`	number of START messages to restart
`#define NRF_WAKEUP`	`(3)`	number of wakeup frames
`#define NRF_SYNC`	`(13)`	length of sync in frames
`#define NRF_OBSERVE`	`(10)`	number of frames the PSAP checks for a better sync after detecting a preamble
`#define PNSEQ_OSF`	`(22)`	"oversampling" rate of PN sequence
`#define PEAK_DIST_PP`	`(30*PNSEQ_OSF)`	distance outer positive peaks
`#define PEAK_DIST_NN`	`(54*PNSEQ_OSF)`	distance negative peaks
`#define PEAK_DIST_PN`	`(12*PNSEQ_OSF)`	distance positive to negative
`/* Uplink/Downlink format */`		
`#define ARQ_MAX`	`(8)`	number of redundancy versions
`#define NRB_TAIL`	`(3)`	number of encoder tail bits
`#define NRB_CRC`	`(28)`	order of CRC polynomial
`#define NRB_INFO`	`(8*MSD_MAX_LENGTH)`	
`#define NRB_INFO_CRC`	`(8*MSD_MAX_LENGTH + NRB_CRC)`	
`#define NRB_CODE_ARQ`	`(1380)`	
`#define NRB_CODE_BUFFER`	`(3*(8*MSD_MAX_LENGTH + NRB_CRC) + 4*NRB_TAIL)`	
`#define SET_LLMSG`	`(16)`	set size lower-layer messages
`#define SET_HLMSG`	`(16)`	set size higher-layer messages

```
#define NRF_DLDATA              (3)            downlink data frames
#define NRF_DLMUTE1LL           (2)            1st muting lower-layer message
#define NRF_DLMUTE1HL           (1)            1st muting higher-layer message
#define NRF_DLCHUNK             (NRF_SYNC + NRF_DLMUTE1HL + 2*NRF_DLDATA)

/* IVS/PSAP processing */
#define NRF_MEMCTRL             (7)            buffer size in frames
#define NRS_MEMSYNC             (820)          memory size in samples (SYNC)

#define IVS_THRESHOLD           (40000)        threshold for control messages
#define IVS_GOSTART             (6)            threshold for unreliable START
#define IVS_TXFAST              (10)           fast modulator mode NACK condition
#define IVS_TXINC               (87)           sample increment at restart

#define PSAP_NUMSTART           (20)           number of START messages
#define PSAP_NUMACK             (20)           number of ACK messages
#define PSAP_NUMHLACK           (5)            number of PSAP HLACK messages
#define PSAP_THRESHOLD          (40)           threshold for modulator type

#define FEC_VAR                 (30206)        variance: 1/4550000 in Q37
#define FEC_MEAN                (0xB9999A)     mean: 5.8 in Q21
#define FEC_ITERATIONS          (8)            number of decoder iterations
#define FEC_STATES              (8)            number of decoder states

#define IntLLR                  Int16          size of soft bit buffer variables
#define LLR_MAX                 ((Int32)(0x7fff-1))
#define LOGEXP_RES              (401)          resolution of LOGEXP table
#define LOGEXP_DELTA            (-6)           determines internal Q-factor
#define LOGEXP_QIN              (8)            input Q-factor of LLR values
```

4.3.2 Type Definitions

The following type definitions have been used, which are defined in ecall_defines.h, ecall_modem.h, ecall_sync.h, and modemx.h:

Definition	Description
`typedef enum { False, True } Bool;`	boolean variable
`typedef enum { Minus = -1, Zero, Plus } Tern;`	ternary variable
`typedef signed char Int8;`	8 bit signed variable
`typedef signed short int Int16;`	16 bit signed variable
`typedef signed int Int32;`	32 bit signed variable
`typedef unsigned char Ord1;`	binary symbol
`typedef unsigned char Ord8;`	8 bit unsigned variable
`typedef unsigned short int Ord16;`	16 bit unsigned variable
`typedef unsigned int Ord32;`	32 bit unsigned variable
`typedef enum { ModUndef, Mod3bit4smp, Mod3bit8smp } ModType;`	modulator type for uplink transmission
`typedef struct { ModType type;`	identifies modulator type
`Int16 bpsym;`	bits per symbol
`Int16 spmf;`	samples per modulation frame
`Int16 mfpf;`	modulation frames per frame = PCM_LENGTH/spmf
`Int16 decpos1;`	position 1st decoding trial
`Int16 decpos2;`	position 2nd decoding trial
`Int16 wutperiod;`	wakeup tone period in samples
`Int16 nfmute1;`	number of muting frames 1st interval
`Int16 nfmute4;`	number of muting frames 4th interval
`Int16 nfmuteall;`	number of muting frames total
`Int16 nfdata;`	number of data frames = NRB_CODE_ARQ/(mfpf*bpsym)

```
  const Int16 *ulPulse;
  const Int16 *ulPulseMatch;
  const Int16 *mgTable;
  const Int16 *wakeupSin;
  const Int16 *wakeupCos;
} ModState;                          modulator state for uplink transmission

typedef struct {
  Int32 *mem;                        memory for sync
  Int32 *memWakeup;                  memory for wakeup tone detector

  Int32 amplitude[3];                amplitudes (average, maximum, memory)
  Int16 corrIndex[4];                sync check positions
  Int32 corrCheck[4];                sync check correlation values
  Int16 peakPos[4];                  sync peak positions within control message

  Bool  flag;                        indicates successful sync
  Int16 delay;                       synchronization delay (position)
  Int16 tempDelay;                   temporary delay in two-stage peak evaluation
  Int16 prevDelay;                   previous sync delay
  Int16 trials;                      number of sync trials
  Int16 npeaks;                      number of sync peaks detected
  Int16 events;                      number of subsequent equal sync events

  Tern  check;                       indicates sync check result (ternary variable)
  Int16 checkCnt;                    counter for subsequent sync check failures
  Int16 index;                       frame reference for sync evaluation
  Int16 offset;                      frame offset
} SyncState;                         state of synchronization functions

typedef enum {
  DlNoop = -2,
  DlTriggerReset,
  DlMsgStart,
  DlMsgNack,
  DlMsgAck,
  DlMsgHlack = SET_LLMSG
} DlData;                            downlink message identifiers

typedef enum {
  IvsIdle,
  IvsTrigger,
  IvsStart,
  IvsSendMsd,
  IvsAck
} IvsState;                          IVS state identifiers

typedef struct {
  CtrlRxData ctrl;                   IVS control struct
  SyncState  sync;                   IVS sync struct

  Int16 state;                       receiver state
  Int16 dlData;                      downlink message symbol
  Int16 dlIndex;                     donwlink frame counter
  Int16 dlMsgCnt;                    downlink message counter

  Int16 memCtrl[NRF_MEMCTRL*PCM_LENGTH];
  Int32 memSync[NRS_MEMSYNC];
} IvsRxData;

typedef struct {
  CtrlTxData ctrl;                   IVS control struct
  ModState mod;                      IVS modulator struct
  Int16 state;                       transmitter state
  Int16 stateCntNack;                global NACK counter
  Bool  startPending;                indicates pending START message

  Int16 delay;                       transmit offset in samples
  Int16 rv;                          redundancy version
```

```
  Int16 ulN;                          uplink number of frames
  Int16 ulIndex;                      uplink frame counter
  Int16 ulDelay;                      uplink transmit offset in samples

  Int16 stateCnt[SET_LLMSG + 1];      state counters
  Int16 stateIgn[SET_LLMSG + 1];      counter for unreliable messages

  Ord1  memCode[NRB_CODE_BUFFER];
  Int16 memDelay[2*PCM_LENGTH];
} IvsTxData;

typedef struct {
  IvsRxData rx;                       IVS receiver struct
  IvsTxData tx;                       IVS transmitter struct
} IvsData;

typedef enum {
  PsapIdle,
  PsapTrigger,
  PsapStart,
  PsapNack,
  PsapAck,
  PsapHlack,
} PsapState;                          PSAP state identifiers

typedef struct {
  CtrlRxData ctrl;                    PSAP control struct
  SyncState sync;                     PSAP sync struct
  ModState  mod;                      PSAP modulator struct

  Int16 state;                        receiver state
  Int16 rv;                           redundancy version
  Int16 ulN;                          uplink number of frames without muting
  Int16 ulIndex;                      uplink frame counter
  Int16 mgIndex;                      uplink position in muting gap table
  Int16 ulTrials;                     uplink decoding trails
  Int16 ulSyncTail;                   sync observation counter after sync success

  Ord8  dlHlackData;                  downlink higher-layer message (4 bits)
  Int16 dlData;                       downlink message symbol
  Int16 dlIndex;                      donwlink frame counter
  Int16 dlMsgCnt;                     downlink message counter

  Ord8   *msd;                        MSD in byte representation
  Ord1   *msdBin;                     MSD in binary representation
  Int16  *memCtrl;                    buffer for control and data demodulation
  IntLLR *memCode;                    soft bit buffer for decoding

  char buffer[0
    + sizeof(IntLLR)* NRB_CODE_ARQ
    + sizeof(Int16) * NRF_MEMCTRL*PCM_LENGTH
    + sizeof(Int32) * NRS_MEMSYNC
    + sizeof(Int32) * 2*(NRF_SYNC+1)];
} PsapRxData;

typedef struct {
  CtrlTxData ctrl;                    PSAP control struct
} PsapTxData;

typedef struct {
  PsapRxData rx;                      PSAP receiver struct
  PsapTxData tx;                      PSAP transmitter struct
  Int16 msgCounter;                   message counter
} PsapData;

typedef enum {
  CtrlRxIdle,
  CtrlRxSync,
  CtrlRxLock,
  CtrlTxIdle,
```

```
  CtrlTxSend
} PortState;

typedef struct {
  Int16 dlData;                       message symbol
  Int16 dlIndex;                      message frame counter
} CtrlTxPort;

typedef struct {
  Int16 dlData;                       detected message symbol
  Int16 dlMetric;                     receiver metric
} CtrlRxPort;


typedef struct {
  CtrlPort port;                      port struct
} CtrlTxData;

typedef struct {
  CtrlPort port;                      port struct

  SyncState *sync;                    pointer to sync struct
  Int16 *buffer;                      pointer to control receiver buffer

  Ord8  dlHlackData;                  downlink higher-layer message (4 bits)
  Tern  dlRead;                       sync indication (ternary variable)
  Int16 dlIndex;                      internal frame counter
  Int16 dlSyncLock;                   number of sync events required
} CtrlRxData;
```

4.3.3 Description of fixed tables used in the C-code

This clause contains a listing of all fixed tables (ROM) defined in `ecall_rom.c`.

Type/Constant	Dimension	Description
/* Synchronization */		
Int16 wakeupSin500	[16]	sine waveform at 500 Hz
Int16 wakeupCos500	[16]	cosine waveform at 500 Hz
Int16 wakeupSin800	[10]	sine waveform at 800 Hz
Int16 wakeupCos800	[10]	cosine waveform at 800 Hz
Int16 syncPulseForm	[5]	sync pulse
Int16 syncSequence	[15]	sync pulse sequence
Int16 syncIndexPreamble	[SYNC_IDXLEN]	sync pulse positions
Int16 syncFrame	[1600]	predefined synchronization signal
/* Uplink/Downlink format */		
Int16 indexBits	[24]	bit positions for turbo decoder
// fast modulator mode:		
Int16 m4smp_ulPulse	[16]	uplink waveform
Int16 m4smp_ulPulseMatch	[64]	matched filtered uplink waveform
Int16 m4smp_mgTable	[54]	table indicating muting gaps
// robust modulator mode:		
Int16 m8smp_ulPulse	[32]	uplink waveform
Int16 m8smp_ulPulseMatch	[128]	matched filtered uplink waveform
Int16 m8smp_mgTable	[104]	table indicating muting gaps
Int16 dlPcmData	[4][NRF_DLDATA*PCM_LENGTH]	downlink transmit signal
Int16 dlPcmDataMatch	[4][NRF_DLDATA*PCM_LENGTH]	DL MF signal
/* FEC encoder/decoder */		
Ord16 stateTransMat	[8][2]	FEC: state transitions
Ord16 stateTrans	[16]	FEC: state transitions
Ord16 revStateTransMat	[8][2]	FEC: reverse state transitions
Ord16 revStateTrans	[16]	FEC: reverse state transitions
Ord1 outputParityMat	[8][2]	FEC: output parity indicator

```
Ord1  outputParity            [16]                    FEC: output parity indicator

Ord1  crcPolynomial           [NRB_CRC+1]             coefficients of CRC polynomial
Ord1  scramblingSeq           [NRB_INFO_CRC]          bit scrambling sequence
Ord16 interleaverSeq          [NRB_INFO_CRC]          interleaver sequence
Ord16 redVerIndex             [8][NRB_CODE_ARQ]       index vector for HARQ process

IntLLR logExpTable            [LOGEXP_RES]            lookup table (logExp function)
```

4.3.4 Static variables used in the C-code

This clause contains a listing of static variables (RAM) defined in source files.

Definition	Description
`IvsData ivs`	`IVS static memory`
`PsapData psap`	`PSAP static memory`
`WordLLR chCodedSoftBitBuffer[NRB_CODE_BUFFER]`	`soft bit buffer of turbo decoder`

4.4 Functions of the C Code

This clause contains the headers of the employed IVS and PSAP functions. They correspond to a large extent to the functional description of the IVS and PSAP provided in 3GPP TS 26.267 [1].

Figure 1 gives an overview of the most important functions and their hierarchical relation.

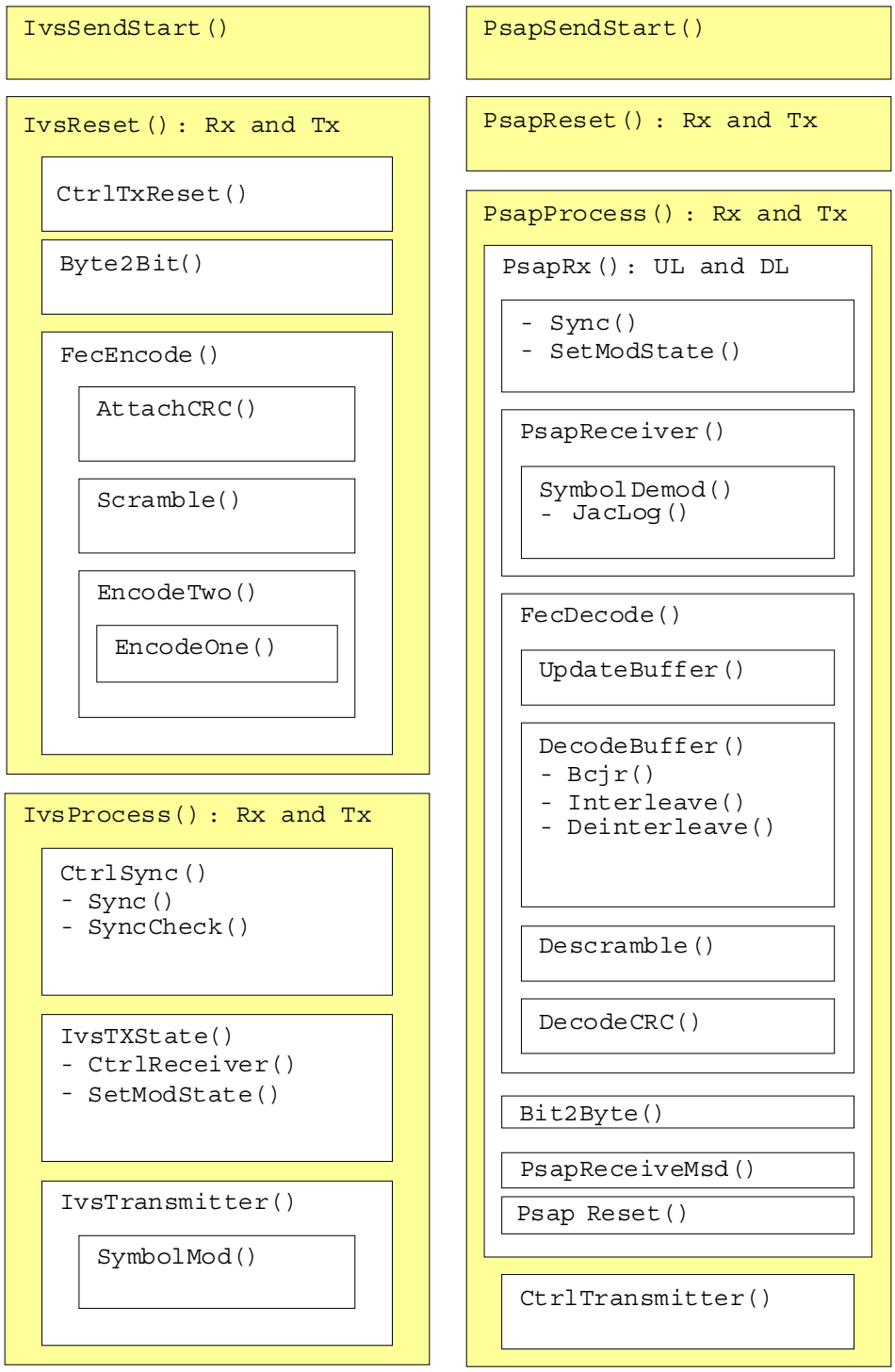


Figure 1: Hierarchical function overview

4.4.1 Interface functions

```
/*============================================================================*/
/* IVS  implementation: IvsReset                                              */
/*----------------------------------------------------------------------------*/
/* Description: Reset of IVS before the reception of a new MSD                */
/*                                                                            */
/* In:  const Ord8* msd     -> MSD to be transmitted                          */
/*      int         length  -> MSD length (equal to MSD_MAX_LENGTH)           */
/*----------------------------------------------------------------------------*/
```

```
void IvsReset(const Ord8 *msd, int length)
void IvsRxReset()
void IvsTxReset(const Ord8 *msd, int length)


/*============================================================================*/
/* IVS  implementation: IvsProcess                                            */
/*----------------------------------------------------------------------------*/
/* Description: IVS modem function that processes the PCM data                */
/*                                                                            */
/* InOut:  Int16* pcm  <-> input and output frame of 16bit PCM samples        */
/*----------------------------------------------------------------------------*/
void IvsProcess(Int16 *pcm)
void IvsRxProcess(const Int16 *pcm)
void IvsTxProcess(Int16 *pcm)


/*============================================================================*/
/* IVS  implementation: IvsSendStart                                          */
/*----------------------------------------------------------------------------*/
/* Description: Initiates IVS to trigger the transmission of SEND messages    */
/*----------------------------------------------------------------------------*/
void IvsSendStart();


/*============================================================================*/
/* IVS  implementation: IvsReceiveAck                                         */
/*----------------------------------------------------------------------------*/
/* Description: callback function indicating a received ACK message           */
/*----------------------------------------------------------------------------*/
void IvsReceiveAck();


/*============================================================================*/
/* IVS  implementation: IvsReceiveHlap                                        */
/*----------------------------------------------------------------------------*/
/* Description: callback function indicating a received higher layer messages */
/*                                                                            */
/* In:   const Ord8 data -> data symbol identifierer                          */
/*----------------------------------------------------------------------------*/
void IvsReceiveHlap(const Ord8 data);


/*============================================================================*/
/* PSAP  implementation: PsapSendStart                                        */
/*----------------------------------------------------------------------------*/
/* Description: Initiates PSAP to trigger the transmission of an MSD          */
/*----------------------------------------------------------------------------*/
void PsapSendStart()


/*============================================================================*/
/* PSAP  implementation: PsapSendHlap                                         */
/*----------------------------------------------------------------------------*/
/* Description: Initiates PSAP to send the higher layer messages              */
/*                                                                            */
/* In:   const Ord8 data -> data symbol identifierer                          */
/*----------------------------------------------------------------------------*/
void PsapSendHlap(const Ord8 data);


/*============================================================================*/
/* PSAP  implementation: PsapReset                                            */
/*----------------------------------------------------------------------------*/
/* Description: Reset of PSAP before the reception of a new MSD               */
/*----------------------------------------------------------------------------*/
void PsapReset()
void PsapRxReset()
void PsapTxReset()
```

```
/*============================================================================*/
/* PSAP  implementation: PsapProcess                                          */
/*----------------------------------------------------------------------------*/
/* Description: PSAP modem function that processes the PCM data               */
/*                                                                            */
/* InOut:  Int16* pcm  <-> input and output frame of 16bit PCM samples        */
/*----------------------------------------------------------------------------*/
void PsapProcess(Int16 *pcm)
void PsapRxProcess(const Int16 *pcm)
void PsapTxProcess(Int16 *pcm)
```

4.4.2 IVS transmitter functions

```
/*============================================================================*/
/* IVS FUNCTION: IvsTransmitter                                               */
/*----------------------------------------------------------------------------*/
/* Description: IVS transmitter function                                      */
/*                                                                            */
/* In:     const ModState* ms      -> modulator struct                        */
/*         const Ord1*     buffer  -> code bit buffer                         */
/*         Int16           rv      -> redundancy version                      */
/*         Int16           index   -> position within uplink frame            */
/* Out:    Int16*          pcm     <- output data                             */
/*----------------------------------------------------------------------------*/
void IvsTransmitter(const ModState *ms, const Ord1 *buffer, Int16 *pcm,
                    Int16 rv, Int16 index)


/*============================================================================*/
/* UTILITY FUNCTION: IvsTxState                                               */
/*----------------------------------------------------------------------------*/
/* Description: IVS state machine evaluating feedback messages                */
/*                                                                            */
/* In:  Int16 msg       -> new control message symbol                         */
/*      Int16 metric    -> receiver metric (-1: ignore symbol)                */
/*      Bool  syncLock  -> indicates sync lock of control receiver            */
/*----------------------------------------------------------------------------*/
void IvsTxState(Int16 msg, Int16 metric, Bool syncLock)


/*============================================================================*/
/* IVS FUNCTION: SymbolMod                                                    */
/*----------------------------------------------------------------------------*/
/* Description: symbol modulator                                              */
/*                                                                            */
/* In:     const ModState* ms      -> modulator struct                        */
/*         Int16           symbol  -> symbol index                            */
/* Out:    Int16*          mPulse  <- modulated output sequence               */
/*----------------------------------------------------------------------------*/
void SymbolMod(const ModState *ms, Int16 symbol, Int16 *mPulse)


/*============================================================================*/
/* IVS FUNCTION: Byte2Bit                                                     */
/*----------------------------------------------------------------------------*/
/* Description: conversion byte vector to bit vector                          */
/*                                                                            */
/* In:     Ord8* in      -> vector of input bytes                             */
/*         Int16 length  -> length of input                                   */
/* Out:    Ord1* out     <- vector of output bits                             */
/*----------------------------------------------------------------------------*/
void Byte2Bit(const Ord8 *in, Ord1 *out, Int16 length)


/*============================================================================*/
/* ENCODER FUNCTION: FecEncode                                                */
```

```
/*-----------------------------------------------------------------------------*/
/* Description: encoding of MSD                                                */
/*                                                                             */
/* InOut:  Ord1 *buffer  <-> takes info bits and returns coded bits            */
/*-----------------------------------------------------------------------------*/
void FecEncode(Ord1 *buffer)


/*=============================================================================*/
/* ENCODER FUNCTION: AttachCrc                                                 */
/*-----------------------------------------------------------------------------*/
/* Description: attaches CRC bits                                              */
/*                                                                             */
/* In:     const Ord1* infoBits     -> input information bits                  */
/* Out:    Ord1*       infoWithCrc  <- bits with CRC attached                  */
/*-----------------------------------------------------------------------------*/
void AttachCrc(const Ord1 *infoBits, Ord1 *infoWithCrc)


/*=============================================================================*/
/* ENCODER FUNCTION: Scramble                                                  */
/*-----------------------------------------------------------------------------*/
/* Description: bit scrambling                                                 */
/*                                                                             */
/* In:     const Ord1* in   -> non scrambled input bit sequence                */
/* Out:    Ord1*       out  <- scrambled output bit sequence                   */
/*-----------------------------------------------------------------------------*/
void Scramble(const Ord1 *in, Ord1 *out)


/*=============================================================================*/
/* ENCODER FUNCTION: EncodeTwo                                                 */
/*-----------------------------------------------------------------------------*/
/* Description: encoding of bit sequence                                       */
/*                                                                             */
/* InOut:  Ord1* codedBits  <-> scrambled bits to coded bits                   */
/*-----------------------------------------------------------------------------*/
void EncodeTwo(Ord1 *codedBits)


/*=============================================================================*/
/* ENCODER FUNCTION: EncodeOne                                                 */
/*-----------------------------------------------------------------------------*/
/* Description: convolutional encoding of each component                       */
/*                                                                             */
/* In:     Int16 encNr       -> component number                               */
/* InOut:  Ord1* codedBits  <-> bits to be encoded                             */
/*-----------------------------------------------------------------------------*/
void EncodeOne(Ord1 *codedBits, Int16 encNr)
```

4.4.3 PSAP receiver functions

```
/*=============================================================================*/
/* UTILITY FUNCTION: PsapRxUplink                                              */
/*-----------------------------------------------------------------------------*/
/* Description: PSAP UL state machine, determines PSAP receiver operation     */
/*              according to the state                                         */
/*                                                                             */
/* In:  const Int16* pcm  -> input frame of 16bit PCM samples                  */
/*-----------------------------------------------------------------------------*/
void PsapRxUplink(const Int16 *pcm)


/*=============================================================================*/
/* UTILITY FUNCTION: PsapRxDownlink                                          */
/*-----------------------------------------------------------------------------*/
/* Description: PSAP DL state machine, determines PSAP transmitter operation  */
```

```
/*              according to the state                                        */
/*------------------------------------------------------------------------------*/
void PsapRxDownlink()


/*==============================================================================*/
/* PSAP FUNCTION: PsapReceiver                                                  */
/*------------------------------------------------------------------------------*/
/* Description: PSAP receiver function (decoding is done outside)               */
/*                                                                              */
/* In:     const ModState* ms        -> modulator struct                        */
/*         const Int16*    pcm       -> input data for demodulation             */
/* Out:    IntLLR*         softBits  <- demodulated soft bit sequence           */
/*------------------------------------------------------------------------------*/
void PsapReceiver(const ModState *ms, const Int16 *pcm, IntLLR *softBits)


/*==============================================================================*/
/* PSAP FUNCTION: SymbolDemod                                                   */
/*------------------------------------------------------------------------------*/
/* Description: symbol demodulator                                              */
/*                                                                              */
/* In:     const ModState* ms        -> modulator struct                        */
/*         const Int16*    mPulse    -> received pulse train                    */
/* Out:    IntLLR*         softBits  <- demodulated soft bit sequence           */
/*------------------------------------------------------------------------------*/
void SymbolDemod(const ModState *ms, const Int16 *mPulse, IntLLR *softBits)


/*==============================================================================*/
/* PSAP FUNCTION: Bit2Byte                                                      */
/*------------------------------------------------------------------------------*/
/* Description: conversion bit vector to byte vector                            */
/*                                                                              */
/* In:     const Ord1* in      -> vector of input bits                          */
/*         Int16       length  -> length of output                              */
/* Out:    Ord8*       out     <- vector of output bytes                        */
/*------------------------------------------------------------------------------*/
void Bit2Byte(const Ord1 *in, Ord8 *out, Int16 length)


/*==============================================================================*/
/* PSAP FUNCTION: MpyLacc                                                       */
/*------------------------------------------------------------------------------*/
/* Description: multiply 32bit number with 16bit number (32bit result)          */
/*                                                                              */
/* In:     Int32 var32  -> 32bit number                                         */
/*         Int16 var16  -> 16bit number                                         */
/* Return: Int32        <- result                                               */
/*------------------------------------------------------------------------------*/
Int32 MpyLacc(Int32 var32, Int16 var16)


/*==============================================================================*/
/* DECODER FUNCTION: FecDecode                                                  */
/*------------------------------------------------------------------------------*/
/* Description: decoding to find the MSD                                        */
/*                                                                              */
/* In:     const IntLLR* in   -> received soft bits                             */
/*         Int16         rv   -> redundancy version                             */
/* Out:    Ord1*         out  <- decoded MSD in binary representation           */
/* Return: Bool               <- result of CRC check                            */
/*------------------------------------------------------------------------------*/
Bool FecDecode(const IntLLR *in, Int16 rv, Ord1 *out)


/*==============================================================================*/
/* DECODER FUNCTION: UpdateBuffer                                               */
/*------------------------------------------------------------------------------*/
/* Description: update channel LLR buffer with new soft bits                    */
```

```
/*                                                                          */
/* In:     const IntLLR* softInBits    -> received soft bits                */
/*         Int16         rv            -> redundancy version                */
/* InOut:  IntLLR*       chLLRbuffer  <-> decoder buffer                    */
/*--------------------------------------------------------------------------*/
void UpdateBuffer(IntLLR *chLLRbuffer, const IntLLR *softInBits, Int16 rv)


/*==========================================================================*/
/* DECODER FUNCTION: DecodeBuffer                                           */
/*--------------------------------------------------------------------------*/
/* Description: decoding of LLR buffer                                      */
/*                                                                          */
/* In:     const IntLLR* syst1    -> RX systematic soft bits                */
/*         const IntLLR* syst2    -> interleaved RX systematic tail bits    */
/*         const IntLLR* parity1  -> RX parity soft bits                    */
/*         const IntLLR* parity2  -> interleaved RX parity soft bits        */
/* Out:    Ord1*         decBits  <- decoded bits                           */
/*--------------------------------------------------------------------------*/
void DecodeBuffer(const IntLLR *syst1, const IntLLR *syst2,
                  const IntLLR *parity1, const IntLLR *parity2, Ord1 *decBits)


/*==========================================================================*/
/* DECODER FUNCTION: Bcjr                                                   */
/*--------------------------------------------------------------------------*/
/* Description: BCJR algorithm                                              */
/*                                                                          */
/* In:     const IntLLR* parity      -> received parity soft bits           */
/* InOut:  IntLLR*       extrinsic  <-> extrinsic information             */
/*--------------------------------------------------------------------------*/
void Bcjr(const IntLLR *parity, IntLLR *extrinsic)


/*==========================================================================*/
/* DECODER FUNCTION: Interleave                                             */
/*--------------------------------------------------------------------------*/
/* Description: Turbo code interleaver                                      */
/*                                                                          */
/* In:     const IntLLR* in   -> input sequence                             */
/* Out:    IntLLR*       out  <- output sequence                            */
/*--------------------------------------------------------------------------*/
void Interleave(const IntLLR *in, IntLLR *out)


/*==========================================================================*/
/* DECODER FUNCTION: Deinterleave                                           */
/*--------------------------------------------------------------------------*/
/* Description: Turbo code deinterleaver                                    */
/*                                                                          */
/* InOut:  IntLLR* inout  <-> input and deinterleaved output sequence       */
/*--------------------------------------------------------------------------*/
void Deinterleave(IntLLR *inout)


/*==========================================================================*/
/* DECODER FUNCTION: Descramble                                             */
/*--------------------------------------------------------------------------*/
/* Description: descrambles decoded bits                                    */
/*                                                                          */
/* InOut:  Ord1* inout  <-> input and output bit sequence                   */
/*--------------------------------------------------------------------------*/
void Descramble(Ord1 *inout)


/*==========================================================================*/
/* DECODER FUNCTION: DecodeCrc                                              */
/*--------------------------------------------------------------------------*/
/* Description: check CRC of decoded bits                                   */
/*                                                                          */
```

```
/* In:     const Ord1* codedBits  -> decoded bit sequence to be checked       */
/* Return: Bool                   <- result of CRC check                      */
/*-----------------------------------------------------------------------------*/
Bool DecodeCrc(const Ord1 *codedBits)


/*=============================================================================*/
/* DECODER FUNCTION: GammaQ                                                    */
/*-----------------------------------------------------------------------------*/
/* Description: compute gamma values for BCJR algorithm                        */
/*                                                                             */
/* In:     Int16         k          -> bit position                            */
/*         Int16         l          -> state                                   */
/*         const IntLLR* parity     -> received parity bits                    */
/*         const IntLLR* extrinsic  -> sum of extrinsic and systematic bits    */
/* Return: IntLLR                   <- value of gamma(k,l)                     */
/*-----------------------------------------------------------------------------*/
IntLLR GammaQ(Int16 k, Int16 l, const IntLLR *parity, const IntLLR *extrinsic)


/*=============================================================================*/
/* UTILITY FUNCTION: JacLog                                                    */
/*-----------------------------------------------------------------------------*/
/* Description: Jacobian logarithm                                             */
/*                                                                             */
/* In:     IntLLR a  -> value one                                              */
/*         IntLLR b  -> value two                                              */
/* Return: IntLLR    <- Jacobian logarithm                                     */
/*-----------------------------------------------------------------------------*/
IntLLR JacLog(Int32 a, Int32 b)
```

4.4.4 PSAP transmitter functions

```
See control link functions.
```

4.4.5 IVS receiver functions

```
See control link functions.
```

4.4.6 Synchronization functions (IVS and PSAP)

```
/*=============================================================================*/
/* FUNCTION: Sync                                                              */
/*-----------------------------------------------------------------------------*/
/* Description: main synchronization function                                  */
/*                                                                             */
/* InOut:  SyncState*   sync    <-> sync struct                                */
/* In:     const Int16* pcm      -> input frame                                */
/*         const char*  caller   -> modem identification                       */
/*-----------------------------------------------------------------------------*/
void Sync(SyncState *sync, const Int16 *pcm, const char *caller)


/*=============================================================================*/
/* UTILITY FUNCTION: CtrlSync                                                  */
/*-----------------------------------------------------------------------------*/
/* Description: control message sync function                                  */
/*                                                                             */
/* InOut:  CtrlRxData*  control  <-> control struct                            */
/* In:     const Int16* pcm       -> input frame of 16bit PCM samples          */


void CtrlSync(CtrlRxData *control, const Int16 *pcm)
```

```
/*============================================================================*/
/* IVS FUNCTION: SyncCheck                                                    */
/*----------------------------------------------------------------------------*/
/* Description: check whether locked sync is still valid                      */
/*                                                                            */
/* InOut:  SyncState*   sync   <-> sync struct                                */
/* In:     const Int16* pcm     -> input frame                                */
/*         const char*  caller  -> modem identification                       */
/*----------------------------------------------------------------------------*/
void SyncCheck(SyncState *sync, const Int16 *pcm, const char *caller)


/*============================================================================*/
/* UTILITY FUNCTION: ToneDetect                                               */
/*----------------------------------------------------------------------------*/
/* Description: tone detection at 500 Hz or 800 Hz                            */
/*                                                                            */
/* InOut:  SyncState*   sync  <-> sync struct                                 */
/* In:     const Int16* pcm    -> input frame                                 */
/*----------------------------------------------------------------------------*/
void ToneDetect(SyncState *sync, const Int16 *pcm)


/*============================================================================*/
/* UTILITY FUNCTION: UpdatePeak                                               */
/*----------------------------------------------------------------------------*/
/* Description: update sync peak position                                     */
/*                                                                            */
/* In:     const Int32* pos   -> vector of positions                          */
/*         const Int32* corr  -> vector of correlation values                 */
/*         Int16        dist  -> distance to be checked                       */
/* Return: Int16              <- updated peak position                        */
/*----------------------------------------------------------------------------*/
Int16 UpdatePeak(const Int32 *pos, const Int32 *corr, Int16 dist)


/*============================================================================*/
/* UTILITY FUNCTION: CheckPosPeaks                                            */
/*----------------------------------------------------------------------------*/
/* Description: check positive sync peaks                                     */
/*                                                                            */
/* InOut:  SyncState*   sync        <-> sync struct                           */
/* In:     const char*  caller       -> modem identification                  */
/*         const Int32* pCorr        -> vector of correlation values          */
/*         Int16        p1           -> peak position p1                      */
/*         Int16        p2           -> peak position p2                      */
/*         Int16        ppPeaks      -> number correct pos/pos distances      */
/*         Int16        npPeaks      -> number correct neg/pos distances      */
/*         Int16        targetDelay  -> target delay if sync successful       */
/*----------------------------------------------------------------------------*/
void CheckPosPeaks(SyncState *sync, const char *caller, const Int32 *pCorr,
                   Int16 p1, Int16 p2, Int16 ppPeaks, Int16 npPeaks,
                   Int16 targetDelay)


/*============================================================================*/
/* UTILITY FUNCTION: CheckNegPeaks                                            */
/*----------------------------------------------------------------------------*/
/* Description: check negative sync peaks                                     */
/*                                                                            */
/* InOut:  SyncState*   sync        <-> sync struct                           */
/* In:     const char*  caller       -> modem identification                  */
/*         const Int32* nCorr        -> vector of correlation values          */
/*         Int16        n1           -> peak position n1                      */
/*         Int16        n2           -> peak position n2                      */
/*         Int16        nnPeaks      -> number correct neg/neg distances      */
/*         Int16        npPeaks      -> number correct neg/pos distances      */
/*         Int16        targetDelay  -> target delay if sync successful       */
```

```
/*-------------------------------------------------------------------------*/
void CheckNegPeaks(SyncState *sync, const char *caller, const Int32 *nCorr,
                   Int16 n1, Int16 n2, Int16 nnPeaks, Int16 npPeaks,
                   Int16 targetDelay)


/*=========================================================================*/
/* UTILITY FUNCTION: SyncReset                                             */
/*-------------------------------------------------------------------------*/
/* InOut:  SyncState* sync      <-> sync struct                            */
/* In:     Int32*     mem        -> pointer to sync memory                */
/*         Int32*     memWakeup  -> pointer to sync wakeup memory         */
/*-------------------------------------------------------------------------*/
void SyncReset(SyncState *sync, Int32 *mem, Int32 *memWakeup)
```

4.4.7 Control link functions

```
/*=========================================================================*/
/* CONTROL FUNCTION: CtrlTxProcess                                         */
/*-------------------------------------------------------------------------*/
/* Description: process function control transmitter                       */
/*                                                                         */
/* InOut:  CtrlTxData* control  <-> control struct                         */
/*         Int16* pcm           <-> frame of 16bit PCM samples             */
/*-------------------------------------------------------------------------*/
void CtrlTxProcess(CtrlTxData *control, Int16 *pcm)


/*=========================================================================*/
/* UTILITY FUNCTION: CtrlTxMod                                             */
/*-------------------------------------------------------------------------*/
/* Description: control message transmitter using prestored sequences      */
/*                                                                         */
/* In:     Int16  symbol  -> lower-layer or higher-layer message symbol    */
/*         Int16  index   -> position within message frame                 */
/* Out:    Int16* pcm     <- output data                                   */
/*-------------------------------------------------------------------------*/
void CtrlTxMod(Int16 *pcm, Int16 symbol, Int16 index)


/*=========================================================================*/
/* CONTROL FUNCTION: CtrlRxProcess                                         */
/*-------------------------------------------------------------------------*/
/* Description: process function control receiver                          */
/*                                                                         */
/* InOut:  CtrlRxData*  control  <-> control struct                        */
/* In:     const Int16* pcm       -> input frame of 16bit PCM samples      */
/*-------------------------------------------------------------------------*/
void CtrlRxProcess(CtrlRxData *control, const Int16 *pcm)


/*=========================================================================*/
/* UTILITY FUNCTION: CtrlRxDemod                                           */
/*-------------------------------------------------------------------------*/
/* Description: control message receiver                                   */
/*                                                                         */
/* In:     const Int16* pcm     -> input PCM buffer                        */
/* Out:    Int16*       metric  <- reliability factor (-1: skip)           */
/* Return: Int16                <- demodulated message                     */
/*-------------------------------------------------------------------------*/
Int16 CtrlRxDemod(const Int16 *pcm, Int16 *metric)


/*=========================================================================*/
/* CONTROL FUNCTION: CtrlTxReset                                           */
/*-------------------------------------------------------------------------*/
/* Description: reset function control transmitter                         */
```

```
/*                                                                           */
/* InOut:  CtrlTxData* control  <-> control struct                           */
/* In:     const char* owner     -> modem identification                     */
/*---------------------------------------------------------------------------*/
void CtrlTxReset(CtrlTxData *control, const char *owner)


/*===========================================================================*/
/* CONTROL FUNCTION: CtrlRxReset                                             */
/*---------------------------------------------------------------------------*/
/* Description: reset function control receiver                              */
/*                                                                           */
/* InOut:  CtrlRxData* control   <-> control struct                          */
/* In:     const char* owner      -> modem identification                    */
/*         SyncState*  sync       -> pointer to sync struct                  */
/*         Int16*      buffer     -> pointer to control receiver buffer      */
/*         Int16       syncLock   -> number of sync events required          */
/*---------------------------------------------------------------------------*/
void CtrlRxReset(CtrlRxData *control, const char *owner,
                 SyncState *sync, Int16 *buffer, Int16 syncLock)
```

4.4.8 Other utility functions (IVS and PSAP)

```
/*===========================================================================*/
/* UTILITY FUNCTION: SetModState                                             */
/*---------------------------------------------------------------------------*/
/* Description: set the modulator state                                      */
/*                                                                           */
/* In:     Int16     modType   -> type of modulator to use                   */
/* InOut:  ModState* ms       <-> modulator struct                           */
/*---------------------------------------------------------------------------*/
void SetModState(ModState *ms, ModType modType)
```

Annex A (informative): Change history

Change history							
Date	**TSG SA#**	**TSG Doc.**	**CR**	**Rev**	**Subject/Comment**	**Old**	**New**
2009-03	43	SP-090201			Approved at TSG SA#43	2.0.0	8.0.0
2009-06	44	SP-090251	0001	1	Correction of a mismatch with 3GPP TS 26.267 concerning synchronization	8.0.0	8.1.0
2009-06	44	SP-090251	0002	1	Correction concerning modulator initialization	8.0.0	8.1.0
2009-06	44	SP-090251	0003	1	Correction of a mismatch with 3GPP TS 26.267 concerning ACK transmission	8.0.0	8.1.0
2009-06	44	SP-090251	0004	1	Extension of eCall test setup to allow conformance testing of ACK messages	8.0.0	8.1.0
2009-06	44	SP-090251	0005	2	Separation of IVS and PSAP transmitter and receiver functions in the C-code	8.0.0	8.1.0
2009-09	45	SP-090565	0006	1	Integration of higher-layer acknowledgement message	8.1.0	8.2.0
2009-09	45	SP-090576	0007		Integration of IVS-initiated signalling option	8.1.0	8.2.0
2009-09	45	SP-090565	0008		Parameter change in eCall test setup	8.1.0	8.2.0
2009-09	45	SP-090565	0009		Update of receiver-transmitter interfaces for conformance testing	8.1.0	8.2.0
2009-09	45	SP-090565	0010		Corrections and bugfixes of the eCall source code	8.1.0	8.2.0
2009-12	46				Version for Release 9	8.2.0	9.0.0

History

Document history		
V9.0.0	January 2010	Publication